

$$f \otimes h_1 = \left[-\frac{1}{\sqrt{2}}, -\frac{2}{\sqrt{2}}, -\frac{2}{\sqrt{2}}, -\frac{2}{\sqrt{2}}, \frac{3}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right]$$

$$\left[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right]$$

$$2 \downarrow$$

$$W_\psi(1, n) = \left\{ -\frac{2}{\sqrt{2}}, -\frac{2}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\}$$

$$W_\varphi(2, n) = f(n) = \{1, 3, 5, 7, 4, 3, 2, 1\}$$

$$W_\varphi(1, n) = \left\{ \frac{4}{\sqrt{2}}, \frac{12}{\sqrt{2}}, \frac{7}{\sqrt{2}}, \frac{3}{\sqrt{2}} \right\}$$

$$\left[\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right]$$

$$2 \downarrow$$

$$\left[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right]$$

$$2 \downarrow$$

$$W_\psi(0, n) = \{-4, 2\}$$

$$W_\varphi(1, n) \otimes h_1 = \left[-2, -4, \frac{5}{2}, 2, \frac{3}{2} \right]$$

$$f \otimes h_0 = \left[\frac{1}{\sqrt{2}}, \frac{4}{\sqrt{2}}, \frac{8}{\sqrt{2}}, \frac{12}{\sqrt{2}}, \frac{11}{\sqrt{2}}, \frac{7}{\sqrt{2}}, \frac{5}{\sqrt{2}}, \frac{3}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right]$$

$$\left[\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right]$$

$$2 \downarrow$$

$$W_\varphi(0, n) = \{8, 5\}$$

$$W_\varphi(1, n) \otimes h_0 = \left[2, 8, \frac{19}{2}, 5, \frac{3}{2} \right]$$